



Profile

AI researcher working on **parameter-efficient adaptation of pretrained vision models** and **learned physical dynamics**, applied to autonomous driving. My PhD established hybrid physics/deep-learning modeling and out-of-distribution generalization for dynamical systems; I now transfer that work to autonomous driving at an open industrial research lab. I favor methods that exploit the structure of a problem and I work on integrating deep learning into existing non-ML systems.

Experience

Valeo.ai Paris, France

► **AI Researcher** Dec 2024 – Present

Perception, motion forecasting and decision making for autonomous driving; adapting large pretrained models and learning at scale

► **Postdoctoral AI Researcher** Apr 2024 – Nov 2024

Corner-case generation for learning robust self-driving cars

Sorbonne Université ISIR, MLIA Paris, France

► **Postdoctoral Researcher** Jul 2023 – Dec 2023

Supervised ongoing research projects; authored a tutorial on physics-aware deep learning

► **PhD Student & Teaching Assistant** Oct 2019 – Jun 2023

Advisors: Patrick Gallinari & Nicolas Baskiotis
Physics-aware deep learning for dynamical systems: hybrid physics/DL modeling, out-of-distribution generalization, continuous-time dynamics

► **Research Intern, Deep Learning** Feb 2019 – Sep 2019

Imputing spatiotemporal data with generative models

Inria Paris *Research Intern, NLP* 2018

Beihang Univ. *Research Intern, Vision* 2015–16

Education

Sorbonne Université fka UPMC (Paris-6) Paris, France

► **PhD, Machine Learning & Deep Learning** 2023

► **MSc2 DAC** *Master Data Science Paris* 2019

Université Paris Cité fka Paris-Diderot (Paris-7) Paris, France

► **MSc1 MPRI** *Parisian Research Master in CS* 2018

► Univ. Diploma, French Language & Civilization 2017

Beihang University Beijing, China

► **BSc, Computer Science** 2016

Technical Skills

ML / DL PyTorch, JAX, Hugging Face, multi-GPU & multi-node training

Coding Python, C/C++, Java, OCaml, Matlab, $\text{\LaTeX}$
Infra Linux, Slurm, Docker, Git, Weights & Biases

Languages

French Bilingual *last exam* ► C1 (2017)

English Full Professional *last exam* ► B2 (2015)

Mandarin Native

Awards & Honors

Accessit for the 2024 AI Thesis Prize of the **French Association for Artificial Intelligence (AFIA)**

Best Paper & Oral, NeurIPS 2025 CCFM Workshop

Top Reviewer, NeurIPS 2023

Service & Teaching

Conference reviewing — NeurIPS 2021–26, ICLR 2023–26, ICML 2022–26, CVPR 2025–26, ICRA 2026, ECML-PKDD 2021, ACM-MM 2021

Workshop reviewing — CCFM @ NeurIPS 2025, ML4PS @ NeurIPS 2022–24, Physics4ML @ ICLR 2023, SynS&ML @ ICML 2023, ROAM @ ECCV 2024

Teaching — Sorbonne Université, Engineering Dept. (UFR 919), 2019–22, in French. Undergraduate: C programming (L1), algorithmics (L2), probability (L3). Graduate: ML research methodology (M2)

Selected Publications

Conference & journal papers * equal contribution

· E. Kirby, A. Boulch, Y.-H. Xu, **Y. Yin**, G. Puy, É. Zablocki, A. Bursuc, S. Gidaris, R. Marlet, F. Bartoccioni, A.-Q. Cao, N. Samet, T.-H. Vu, and M. Cord. Driving on registers. In **CVPR 2026**.

· Y.-H. Xu*, **Y. Yin***, T.-H. Vu, A. Boulch, É. Zablocki, and M. Cord. PPT: Pre-training with pseudo-labeled trajectories for motion

forecasting. In *ICRA 2026*.

- **Y. Yin**, S. Venkataramanan, T.-H. Vu, A. Bursuc, and M. Cord. IPA: An information-reconstructive input projection framework for efficient foundation model adaptation. *TMLR*, 2025.

Best Paper & Oral, NeurIPS 2025 CCFM WS

- L. Le Boudec, E. de Bézenac, L. Serrano, R. D. Regueiro-Espino, **Y. Yin**, and P. Gallinari. Learning a neural solver for parametric PDE to enhance physics-informed methods. In *ICLR 2025*.
- A. Kassai Koupaï, J. Mifsut-Benet, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. GEPS: Boosting generalization in parametric PDE neural solvers through adaptive conditioning. In *NeurIPS 2024*.
- E. Le Naour, L. Serrano, L. Migus, **Y. Yin**, G. Agoua, N. Baskiotis, P. Gallinari, and V. Guigue. Time series continuous modeling for imputation and forecasting with implicit neural representations. *TMLR*, 2024.
- L. Serrano, L. Le Boudec, A. Kassai Koupaï, T. X. Wang, **Y. Yin**, J.-N. Vittaut, and P. Gallinari. Operator learning with neural fields: Tackling PDEs on general geometries. In *NeurIPS 2023*.
- **Y. Yin***, M. Kirchmeyer*, J.-Y. Franceschi*, A. Rakotomamonjy, and P. Gallinari. Continuous PDE dynamics forecasting with implicit neural representations. In *ICLR 2023*. **Spotlight**
- M. Kirchmeyer*, **Y. Yin***, J. Donà, N. Baskiotis, A. Rakotomamonjy, and P. Gallinari. Generalizing to new physical systems via context-informed dynamics model. In *ICML 2022*. **Spotlight**
- **Y. Yin**, I. Ayed, E. de Bézenac, N. Baskiotis, and P. Gallinari. LEADS: Learning dynamical systems that generalize across environments. In *NeurIPS 2021*.
- **Y. Yin***, V. Le Guen*, J. Donà*, E. de Bézenac*, I. Ayed*, N. Thome, and P. Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In *ICLR 2021*. **Oral; also in J. Stat. Mech. Theory Exp.**
- D. Huang, R.K. Zhang, **Y. Yin**, Y.D. Wang, and Y.H. Wang. Local feature approach to dorsal hand vein recognition by centroid-based circular key-point grid and fine-grained matching. *Image Vis. Comput.*, 2017.

Workshop papers

- **Y. Yin**, P. Khayatan, É. Zablocki, A. Boulch, and M. Cord. ReGentS: Real-world safety-critical driving scenario generation made stable.

In *ECCV 2024 Workshop on W-CODA*.

- L. Serrano, L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. INFINITY: Neural field modeling for reynolds-averaged navier-stokes equations. In *ICML 2023 Workshop on SynS & ML*.
- L. Migus, **Y. Yin**, J. A. Mazari, and P. Gallinari. Multi-scale physical representations for approximating PDE solutions with graph neural operators. In *ICLR 2022 Workshop on GTRL*.
- **Y. Yin**, A. Pajot, E. De Bézenac, and P. Gallinari. Unsupervised inpainting for occluded sea surface temperature sequences. In *CI 2019*.

Preprints

- **Y. Yin**, E. Ramzi, M. Lafon, V. Charraut, V. Bares, Y.-H. Xu, É. Zablocki, A. Boulch, T. Buhet, A. Bursuc, and M. Cord. Pictura: Perspective-view self-play at scale for driving. *arXiv* 2607.26005, 2026.
- Y.-H. Xu, É. Zablocki, **Y. Yin**, E. Ramzi, E. Kirby, A. Boulch, and M. Cord. TOAD: Test-time trajectory optimization for autonomous driving. *arXiv* 2606.07170, 2026.

Full list of publications on [Google Scholar](#)

Selected Talks

Oral, NeurIPS 2025 CCFM Workshop	Dec 2025
Poster, ECCV 2024	Sep 2024
Workshop on Mathematical Foundations of AI, DATAIA-SCAI	Jan 2024
Tutorial, ECML-PKDD 2023	Sep 2023
Seminar, Signal Processing Lab (LTS4), EPFL	May 2023
Spotlight, ICLR 2023	May 2023
AI4Science Talks, Univ. of Stuttgart & NEC Labs Europe	Apr 2023
Seminar, Criteo AI Lab	Nov 2022
Spotlight, ICML 2022	Jul 2022

Full list of talks on [my website](#)